Mahesh Babu

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PROFESSIONAL SUMMARY

AWS DevOps Engineer with **4 years** of experience in CI/CD automation, cloud infrastructure, and containerized environments. Proficient in Jenkins, Docker, Kubernetes, and AWS, with a strong focus on reducing deployment time and automating workflows. Experienced in building and maintaining high-availability systems across **robotics industrial automation, and defense sectors**. Adept at supporting AI-driven workflows, including computer vision data annotation.

TECHNICAL SKILLS

Operating Systems: Linux (Ubuntu, RHEL), Windows
Cloud Platforms: AWS (EC2, ALB, RDS, ECR, EFS, ElastiCache, DocumentDB, CloudWatch, EKS)
CI/CD: Jenkins, GitHub Actions
Version Control: Git, GitHub
Containerization & Registry: Docker, Harbor, Portainer
Container Orchestration: Kubernetes (K8s), Rancher
Security: Trivy, CodeAnt AI, HashiCorp Vault
Monitoring: Prometheus, Grafana
Programming & Scripting: Python, Bash, YAML
Databases & Caching: MongoDB, PostgreSQL, Redis
AI Tools & Data Annotation: Claude AI, Cursor, LabelMe, CVAT
Project Management & Collaboration: Jira, Tuleap, Slack

PROFESSIONAL EXPERIENCE

AWS DevOps Engineer **Sep 2025 - Present**
Ati Robotics Pvt Ltd (R&D) *Bangalore, India*

- Designed and implemented an **Active-Active High Availability** architecture for the Fleet Manager application for an Amazon client on AWS using EC2, ALB, EFS, RDS, DocumentDB, and ElastiCache, achieving failover in **under 30 seconds** through multi-AZ deployment, health checks, and failover testing.
- Managed production Kubernetes workloads on **AWS EKS** through **Rancher**, handling application deployments, scaling, pod troubleshooting, secrets/configurations, and SSL certificate renewals.
- Provisioned and managed **PostgreSQL, MongoDB, and Redis** databases for Fleet Manager (FM), Deployment Manager (DM), and AtiFlow applications across Development and Production environments, handling configuration, access management, and database connectivity.
- Managed database **backup and restore** activities, monitored database health and resource utilization using Prometheus and Grafana, and troubleshot database connectivity and application performance issues.
- Built and maintained parameterized Jenkins CI/CD pipelines for Mule ARM builds and automated vehicle deployments to **AMR bots** using configurable IP addresses and vehicle types, integrating AWS ECR, RBAC, build locking, and Slack notifications.
- Implemented GitHub Actions CI/CD workflows across Fleet Manager, Deployment Manager, and Sanjaya repositories, automating Docker image builds, version tagging, and publishing to Harbor (Dev) and AWS ECR (Staging).
- Developed GitHub Actions-based pull request workflows to automate backend and frontend test execution, generate code coverage reports, and integrate CodeAnt AI for automated code quality analysis.
- Implemented a two-tier **AI-powered PR review pipeline using Claude AI**, integrating static code analysis and semantic code reviews to identify code quality, logic, security, and documentation issues before merging.
- Optimized Mule Docker images through multi-stage builds, dependency cleanup, and Dockerfile improvements to reduce image size and improve build performance.
- Eliminated hardcoded credentials across codebases and Docker configs by migrating to **HashiCorp Vault**, drastically improving secret rotation and access control.
- Integrated Jenkins with Prometheus and Grafana to monitor CI/CD pipelines, track build and deployment performance, and identify operational issues.
- Supported client deployments of new software versions, monitored AMR bot status through the Fleet Manager UI, troubleshot production issues, and tracked and resolved issues through Jira tickets.

DevOps & Computer Vision Engineer **Aug 2023 - Aug 2025**
Raygain Technologies Pvt Ltd (Client: DRDO/CAIR, R&D) *Bangalore, India*

- Designed and maintained Jenkins pipelines to automate end-to-end build, test, and deployment of ROS2 applications in **offline containerized environments**, optimizing Dockerfiles and configuring a local Docker registry for efficient image storage and distribution.
- Annotated images using LabelMe and CVAT for object detection, segmentation, and classification, and deployed ML models on Kubernetes to automate annotation, reducing manual intervention and improving data quality.
- Implemented Prometheus to monitor ROS2 nodes, topics, and system metrics, and configured Grafana dashboards to visualize performance trends and resource usage for early issue detection.
- Configured NGINX as a reverse proxy with HTTPS for Tuleap and Jenkins using self-signed and internal CA certificates, ensuring secure access over private networks in offline environments.
- Managed ROS2 source code repositories and Git branching strategies, ensuring version control and collaboration across distributed teams.
- Utilized Tuleap Community Edition for project tracking and agile practices, integrated with Jenkins for automated CI/CD workflows enabling streamlined deployments and issue tracking.
- Integrated Tuleap and Jenkins with a Zentyal/SOGo mail server to automate pipeline status alerts and project notifications.
- Assisted scientists in understanding DevOps workflows and computer vision processes, and prepared clear documentation for effective usage of DevOps and annotation tools.

DevOps Engineer (Contract) **Jul 2021 - Jul 2022**
Ravina Technologies Pvt Ltd *Bangalore, India*

- Supported CI/CD pipeline automation using Jenkins and Git to streamline application build and deployment processes.
- Managed Linux package installation, updates, and security patching to maintain stable server environments.
- Automated server configuration and package management using **Ansible playbooks**, ensuring consistent and repeatable deployments.
- Managed Linux user accounts, permissions, and Active Directory group access following role-based access control (RBAC) practices.
- Monitored Linux and Windows servers by tracking CPU, memory, disk utilization, and system processes to ensure system health.
- Provisioned and managed AWS EC2 instances and related cloud resources for application hosting and testing.
- Created and managed IAM users, groups, and policies using the AWS Management Console to control access to cloud resources.
- Assisted in deploying and monitoring applications on Kubernetes clusters hosted on AWS, troubleshooting issues using Prometheus and Grafana.

EDUCATION

Bachelor of Technology (B.Tech) in Mechanical Engineering **2017 - 2020**
BVC College of Engineering, Rajahmundry, India

CERTIFICATIONS

- Artificial Intelligence Fundamentals - IBM, 2026
- Introduction to Model Context Protocol - Anthropic, 2026
- AWS Cloud Practitioner (Course) - Udemy, Stephane Maarek, 2026
- DevOps Certification - Star Agile Institute, Bangalore, 2023
- DevOps Fundamentals - IBM Developer Skills Network, 2023
- AWS and DevOps Certification - SDLC Training Institute, Bangalore, 2022